Retail Sales and Data Quality Assessment

Data Quality Test Cases

Version 1.0

Date: August 2026

Test Case ID	Test Area	Test Description	Expected Result	Actual Result	Status
DQ-001	Completeness	Validate required fields for missing values	No expected null values	0 null value identified in the required fields tested	PASS
DQ-002	Row ID Uniqueness	Check for duplicate Row.ID values	No expected duplicate Row IDs	0 duplicate Row IDs among 51,290 records	PASS
DQ-003	Date Validity	Verify Order Date and Ship Date fall under expected analysis period	All dates should be valid and fall within the dataset's expected period	Order date: 2011-01-01 to 2014-12-31 and ship date: 2011-01-03 to 2015-01-07	PASS, provided the dates were valid and the analysis period was defined as shown
DQ-004	Shipping Sequence	Verify Ship Date not earlier than Order Date	Ship Date ≥ Order Date	0 records have a ship date earlier than order date	PASS
DQ-005	Sales Validation	Identify negative sales values	Sales should be ≥ $0, unless negative sales are an approved business scenario	0 records have negative sales value	PASS
DQ-006	Zero Sales	Identify transactions with zero sales	Zero-sales record should be investigated	1 record has sales of $0, quantity of 1, and profit of -1.11$.	FLAGGED

DQ-007	Profit Validation	Identify negative-profit transaction	Negative-profit required business validation as negative profit is an identified condition, not data-quality failure	12,544 records have negative profit	FLAGGED
DQ-008	Market Consistency	Compare Market & Market2 classifications	Geographic mapping should be consistent	10,378 mismatches:9, 994 US → North America; 384 Canada → North America	FLAGGED

QA Findings Requiring Business Clarification

1. Zero Sales (DQ-006)

   Actual Result: 1 record identified with $0 sales, quantity of 1 and negative profit of $1.11.

   Status: Flagged

2. Negative Profit (DQ-007)

   Expected Result:

   Negative profit transactions should be identified and reviewed to determine whether they represent valid business scenarios.

   Actual Result: 12,544 records contain negative profit

   Status: Flagged

3. Market Consistent (DQ-008)

   Expected Result: Difference between Market and Market2 should be understood and consistent with the intended geographic hierarchy.

   Actual Result: 10,378 mismatches identified: 9,994 US → North America; 384 Canada → North America

   Status: Flagged